



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SAVEETHA SCHOOL OF ENGINEERING, SIMATS
Department of Computer Science and Engineering
Assignment Information

Particular	Details
Name:	K.Dhilip
Register Number:	192365013
Department:	B.E.CSE (Cyber Security)
Programme:	CSE
Course Code & Course Name	CSA1522 – Cloud Computing and Big Data Analytics
Academic Year / Batch	2026–27
Faculty Name	Dr P Rajaram
Assignment Title	Cloud-Based Placement Data Management and Big Data Analytics using JOHO
Date of Issue	31-08-2026
Date of Submission	01-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Understand the cloud services with different service models. CO2: Demonstrate the deployment of Virtualization and build a data center using virtualization tools. CO3: Build methods to access cloud and various techniques to collaborate within cloud.
Bloom's Taxonomy Level	L3 – Apply and L4 – Analyse
SDG Mapping	SDG 4 – Quality Education SDG 8 – Decent Work and Economic Growth SDG 9 – Industry Innovation and Infrastructure SDG 11 – Sustainable Cities and Communities
Industry	Applies cloud services, virtualization and distributed big-data processing to a ZOHO-based placement workflow, supporting scalable access, collaboration and analytics while considering security, privacy and responsible data use.

Problem Statement:

Design and evaluate a simple cloud-based placement data management and access solution for a university using the ZOHO platform. The placement cell needs a reliable way to organize placement information and provide controlled access to students and placement staff. Students shall select suitable IaaS, PaaS and SaaS service models, design a basic cloud architecture, create and configure virtual machines using Oracle VM VirtualBox, demonstrate secure access to a cloud-hosted web application or Web API, and show how users can collaborate through shared cloud resources.

Assessment Rubrics

Assessment Criterion / Marks	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
CO1 – Cloud Service Models & Architecture 25 Marks	Correctly identifies the JOHO requirements, selects appropriate IaaS/PaaS/SaaS models, and presents a complete architecture with clear justification.	Selects appropriate service models and presents a suitable architecture with minor gaps in justification or detail.	Identifies basic service models and provides a simple architecture, but the justification or mapping to requirements is limited.	Service models are unsuitable or misunderstood; architecture is missing/unclear and justification is inadequate.
CO2 – Virtualization & VM Deployment 25 Marks	Creates and configures two VMs correctly using Oracle VM VirtualBox; records CPU, memory, storage and networking; demonstrates communication and clearly explains the hypervisor.	Creates and configures both VMs successfully with mostly correct resources and networking; communication works with minor issues.	Creates the VMs and demonstrates basic operation, but configuration, networking, testing or explanation is incomplete.	VM creation/configuration is incorrect or incomplete; communication is not demonstrated and supporting evidence is insufficient.
CO3 – Cloud Access & Collaboration 25 Marks	Demonstrates secure cloud access successfully, including authentication/access control, clear request-response evidence and a practical collaboration workflow.	Demonstrates working cloud access and collaboration with minor gaps in security, explanation or evidence.	Demonstrates basic cloud access, but collaboration, authentication or security evidence is limited.	Cloud access or collaboration is non-functional, unclear, insecure or not supported by evidence.
Analysis, Validation & Engineering Justification 15 Marks	Compares feasible alternatives using clear criteria, performs relevant tests, interprets results accurately and gives a well-supported final decision.	Provides a reasonable comparison and testing with generally correct interpretation and justification.	Provides limited comparison or testing; interpretation and justification are basic.	Little or no meaningful comparison, testing, interpretation or engineering justification is provided.
Security, Privacy, Ethics & SDG Relevance 10 Marks	Clearly addresses security, privacy, ethical handling of placement data and relevant SDGs with specific application to the JOHO scenario.	Addresses relevant security, privacy, ethical and SDG considerations with minor omissions.	Mentions general security, privacy, ethics or SDG points but gives limited connection to the problem.	Important security, privacy, ethical or SDG considerations are missing or inappropriate.

Deliverables

To receive full credit, each submission must include the following deliverables:

1. Pseudo code
2. Implementation & Results
3. Github and GCR Upload

Assignment Problem / Challenge

The university placement cell requires a simple and reliable system to manage placement-related information such as student details, company details, placement drives and placement results. The challenge is to provide cloud-based access to this information while maintaining controlled access and enabling collaboration between students and placement staff.

The solution was implemented using Zoho Creator as the cloud application platform. The application provides centralized placement data management, reports and dashboards, controlled user access and shared access to placement information.

To demonstrate virtualization and networking concepts, two virtual machines, Placement-Server and Placement-Client, were created using VMware Workstation Pro. Both virtual machines were configured with CPU, memory, storage and a Host-only network. Communication between the two VMs was tested using the `ping` command, and successful two-way communication was achieved with 0% packet loss.

The assignment therefore focuses on applying cloud service models, cloud application development, access control, collaboration, virtualization and virtual networking concepts to a university placement management scenario.

Problem Statement

The university placement cell requires a cloud-based placement management solution to organize student, company, placement drive and placement result information. The system should provide controlled access to students and placement staff and support sharing of placement information through a common cloud platform.

The solution is developed using Zoho Creator for placement data management, secure access and collaboration. In addition, VMware Workstation Pro is used to create and configure two virtual machines, Placement-Server and Placement-Client, and demonstrate network communication between them using a Host-only network.

The proposed solution demonstrates the practical use of SaaS, PaaS, IaaS, cloud computing, access control, collaboration, virtualization and networking concepts in a university placement scenario.

Requirements and Constraints

Functional Requirements

- Manage student, company, placement drive and placement result details.
- Provide controlled access to students and placement staff.
- Support sharing and collaboration through the cloud application.
- Create two virtual machines and test communication between them.

Technical Requirements

- **Zoho Creator** for cloud-based placement management.
- **VMware Workstation Pro** for virtualization.

- **Ubuntu Linux** for both VMs.
- **Host-only network** for VM communication.

Security & Privacy

- Provide controlled access to placement information.
- Prevent unauthorized access to user data.

Reliability

- Placement information should be centrally stored and easily accessible.
- Both VMs should communicate successfully over the network.

Constraints

- Limited CPU, RAM and storage resources for virtual machines.
- Simple implementation suitable for academic demonstration.
- Internet/cloud platform dependency for Zoho Creator.

Student Work

Problem Understanding and Formulation

What is the problem?

The university placement cell needs a centralized cloud-based system to manage student, company, placement drive and placement result information with controlled access for students and placement staff.

Expected Outcomes

- A working cloud-based placement management application.
- Organized placement information and reports.
- Controlled user access and collaboration.
- Two configured virtual machines.
- Successful communication between the virtual machines.

Information/Data Available

The application contains:

- Student details
- Company details
- Placement drive details
- Placement result details
- Reports and dashboard information

Assumptions

- Users have internet access to use the cloud application.
- Students and placement staff have appropriate access permissions.

- The computer has sufficient resources to run two virtual machines.

Constraints

- Limited CPU, RAM and storage resources.
- Simple implementation suitable for academic demonstration.
- Secure and controlled access to placement information.

Application of Course Knowledge

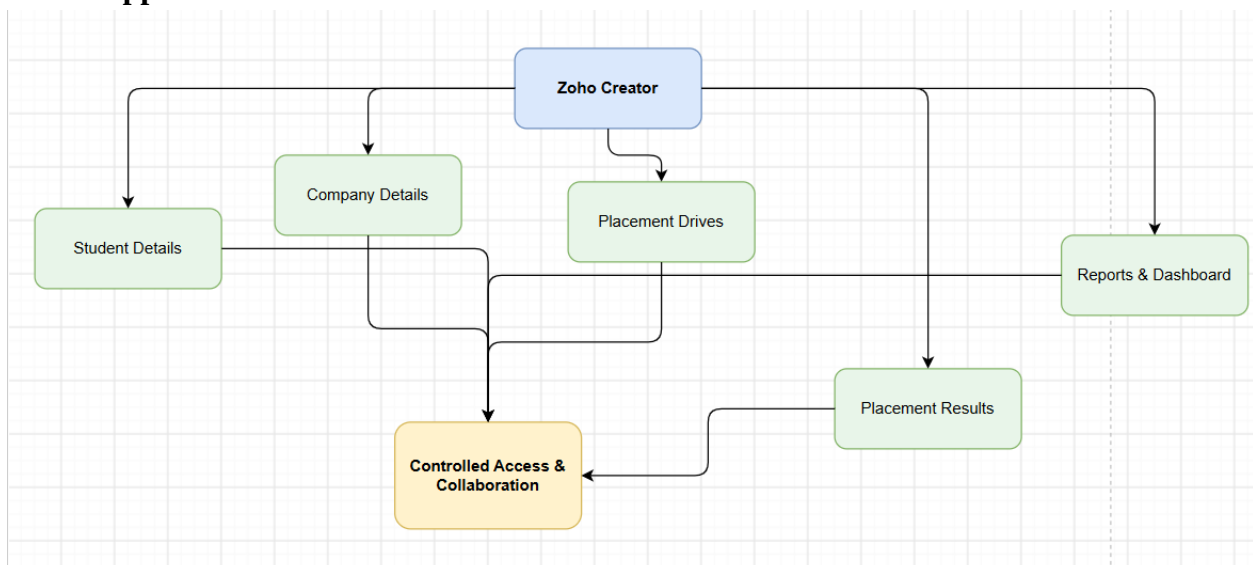
The following cloud computing concepts were applied:

- **SaaS:** Zoho Creator application is accessed through the cloud.
- **PaaS:** Zoho Creator provides a platform for developing the placement application.
- **IaaS:** Virtualized computing resources were demonstrated using VMware virtual machines.
- **Virtualization:** Two Ubuntu VMs were created as Placement-Server and Placement-Client.
- **Networking:** Host-only networking was used for VM-to-VM communication.
- **Security:** Controlled access was applied to placement information.
- **Cloud Collaboration:** Students and placement staff can access and share placement information through the cloud application.

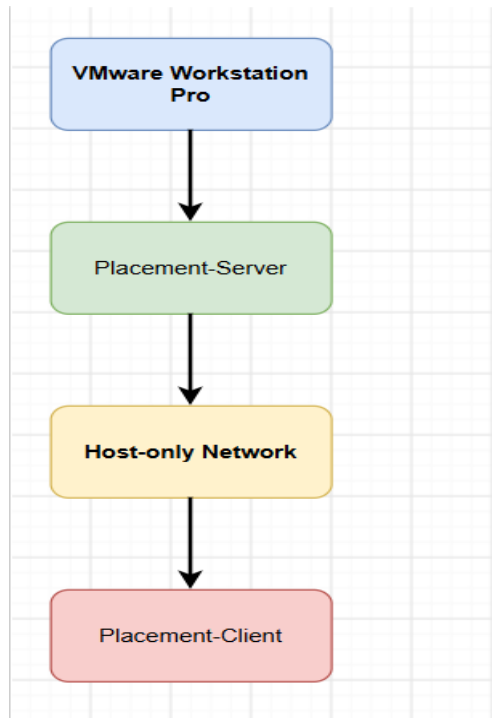
Solution / Design / Methodology

The proposed solution consists of two main parts.

Cloud Application



Virtualization



Both VMs were configured and their communication was tested using the ping command.

Use of Modern Tools

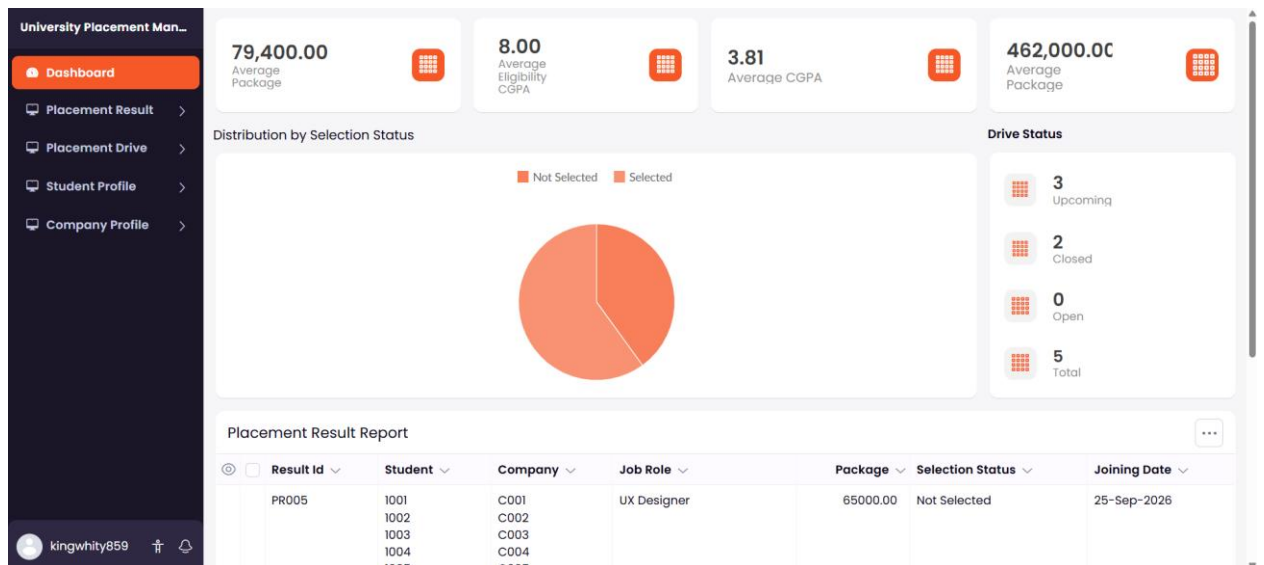
Tool	Purpose
Zoho Creator	Cloud-based placement application
VMware Workstation Pro	Virtual machine creation and configuration
Ubuntu Linux	Operating system for both VMs
Web Browser	Accessing Zoho Creator
Ping	Testing VM network communication

Screenshots of the Zoho application, VM configuration, IP addresses and ping results are provided as evidence.

Results and Validation

The developed system successfully managed placement information through Zoho Creator and provided cloud-based access and collaboration.

1. Zoho Creator placement dashboard.



2. Student Profile module.

Student Profile

Student id: 1

Student Name: Dhillip
First Name

Email: dk@gmail.com

Mobile Number: +91 9123456789

Department: CSE

Year: 4th Year

CGPA: 8.86

Graduation Year: 2027

Skills: I know everything

Placement Status: Not Placed

3. Company Profile module.

Company Profile

Company Id: 1

Company Name: TCS

Industry: Production Sector

Location: Delhi

HR Name: Subahan
First Name

HR Email: sub@gmail.com

Job Role: HR

Package: 70000

Job Description: YES

4. Placement Drive module.

The screenshot shows the 'Placement Drive' module in the 'University Placement Man...' system. The left sidebar contains a navigation menu with options: Dashboard, Placement Result, Placement Drive (selected), Placement Drive R..., Eligible Department, Student Profile, and Company Profile. The main content area is titled 'Placement Drive' and contains a form with the following fields: Drive Id (1), Company (C001), Job Role (HR), Drive Date (04-Sep-2026), Registration Deadline (02-Sep-2026), Eligibility CGPA (7.5), Eligible Department (CSE), Required Skills (Full stack), and Number Of Vacancies (2). At the bottom of the sidebar, the user 'kingwhity859' is logged in.

5. Placement Result module.

The screenshot shows the 'Placement Result' module in the 'University Placement Man...' system. The left sidebar contains a navigation menu with options: Dashboard, Placement Result (selected), Placement Drive, Placement Result R..., Student Profile, and Company Profile. The main content area is titled 'Placement Result' and contains a form with the following fields: Result Id (1), Student (1001), Company (C001), Job Role (HR), Package (45000), Selection Status (radio buttons for Not Selected, Selected (checked), and Other), and Joining Date (10-Sep-2026). At the bottom of the form are 'Submit' and 'Reset' buttons. At the bottom of the sidebar, the user 'kingwhity859' is logged in.

Two virtual machines were successfully configured:

VM	IP Address
Placement-Server	192.168.204.128
Placement-Client	192.168.204.129

Network Validation

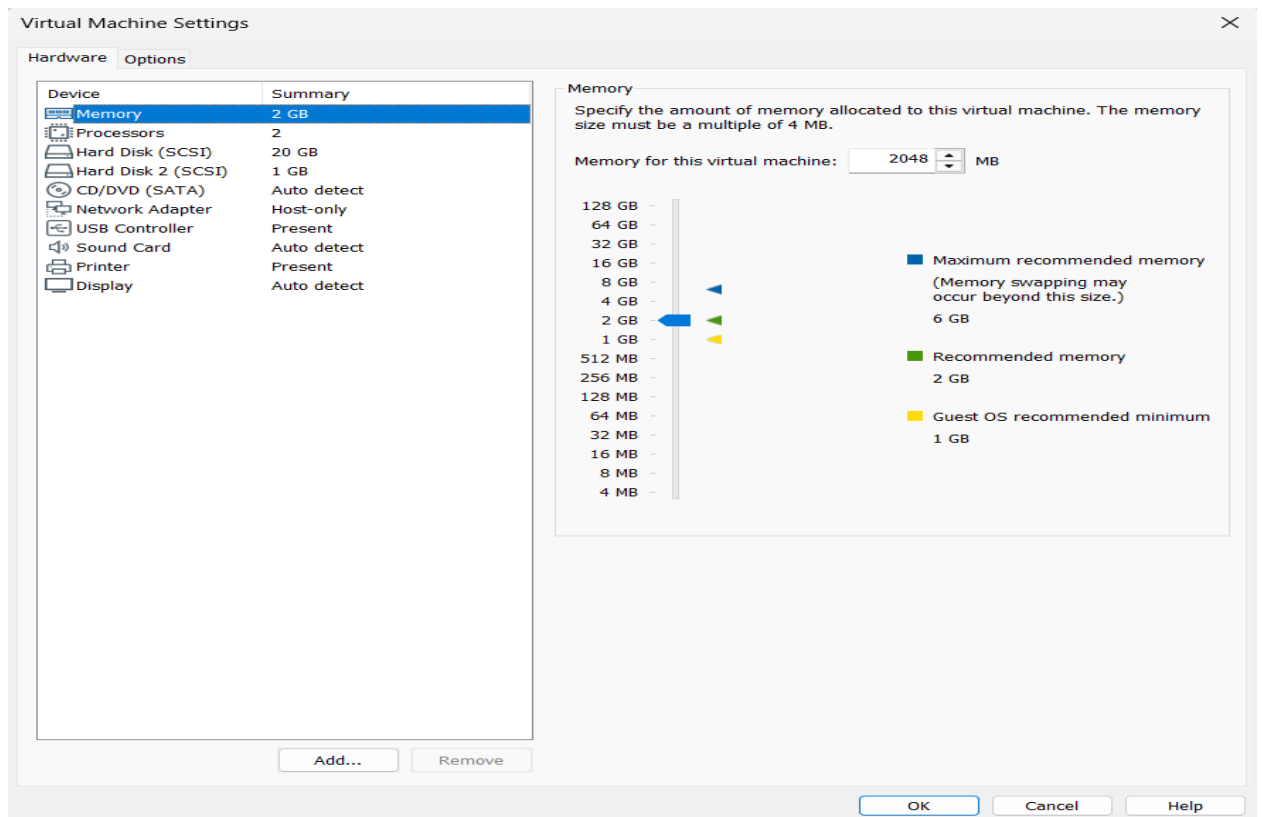
Client → Server:

4 packets transmitted
4 packets received
0% packet loss

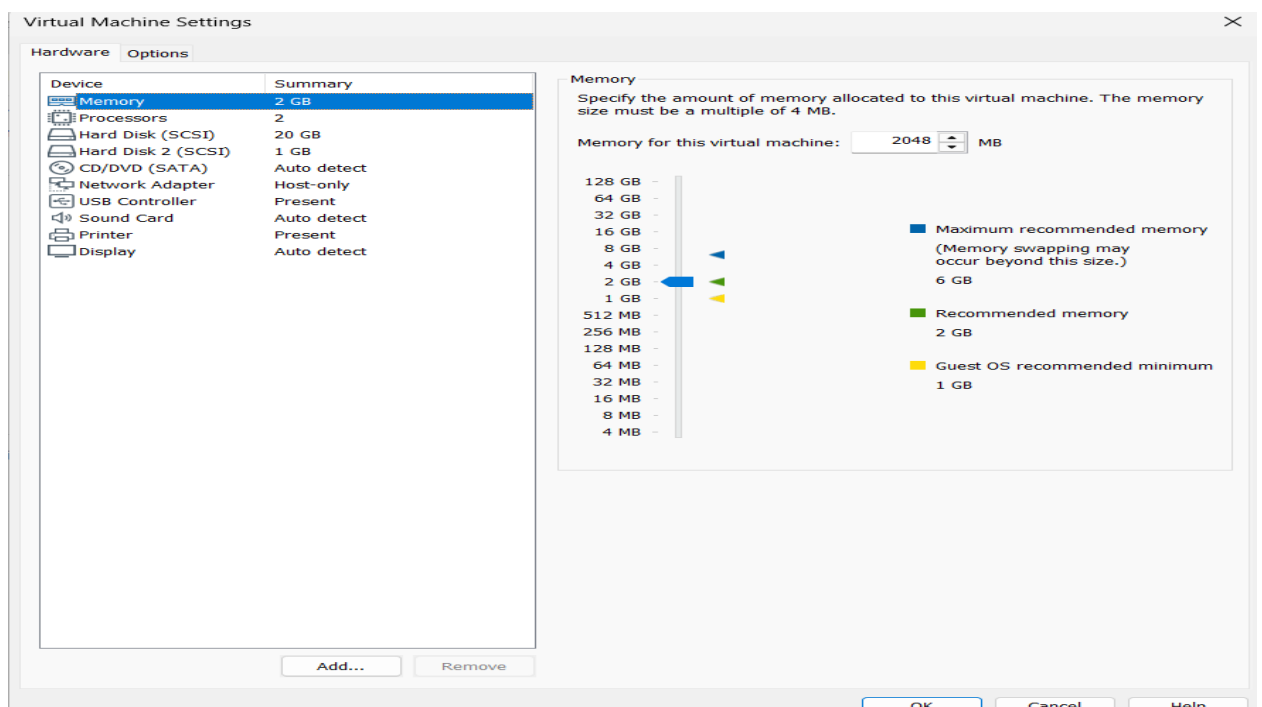
Server → Client:

4 packets transmitted
4 packets received
0% packet loss

1. VMware Placement-Server configuration.



2. VMware Placement-Client configuration.



3. Placement-Server IP address.

```
dhilip@Placement-Server: ~  
File Edit View Search Terminal Help  
valid_lft forever preferred_lft forever  
inet6 ::1/128 scope host  
valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro  
up default qlen 1000  
link/ether 00:0c:29:7d:3c:87 brd ff:ff:ff:ff:ff:ff  
inet 192.168.204.128/24 brd 192.168.204.255 scope global dynamic noprefixrou  
te ens33  
valid_lft 1747sec preferred_lft 1747sec  
inet6 fe80::666c:9e7e:7e33:d290/64 scope link noprefixroute  
valid_lft forever preferred_lft forever
```

4. Placement-Client IP address.

```
dhilip@Placement-Client: ~  
File Edit View Search Terminal Help  
dhilip@Placement-Client:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
inet 127.0.0.1/8 scope host lo  
valid_lft forever preferred_lft forever  
inet6 ::1/128 scope host  
valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
link/ether 00:0c:29:5c:07:f3 brd ff:ff:ff:ff:ff:ff  
inet 192.168.204.129/24 brd 192.168.204.255 scope global dynamic noprefixroute ens33  
valid_lft 1732sec preferred_lft 1732sec  
inet6 fe80::4629:c605:110a:572a/64 scope link noprefixroute  
valid_lft forever preferred_lft forever
```

5. Client-to-Server successful ping.

```
dhilip@Placement-Server:~$ ping -c 4 192.168.204.129  
PING 192.168.204.129 (192.168.204.129) 56(84) bytes of data.  
64 bytes from 192.168.204.129: icmp_seq=1 ttl=64 time=4.13 ms  
64 bytes from 192.168.204.129: icmp_seq=2 ttl=64 time=1.71 ms  
64 bytes from 192.168.204.129: icmp_seq=3 ttl=64 time=3.06 ms  
64 bytes from 192.168.204.129: icmp_seq=4 ttl=64 time=1.61 ms  
  
--- 192.168.204.129 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3007ms  
rtt min/avg/max/mdev = 1.612/2.630/4.134/1.040 ms
```

6. Server-to-Client successful ping.

```
dhilip@Placement-Client:~$ ping -c 4 192.168.204.128  
PING 192.168.204.128 (192.168.204.128) 56(84) bytes of data.  
64 bytes from 192.168.204.128: icmp_seq=1 ttl=64 time=4.31 ms  
64 bytes from 192.168.204.128: icmp_seq=2 ttl=64 time=1.81 ms  
64 bytes from 192.168.204.128: icmp_seq=3 ttl=64 time=2.24 ms  
64 bytes from 192.168.204.128: icmp_seq=4 ttl=64 time=1.08 ms  
  
--- 192.168.204.128 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3005ms  
rtt min/avg/max/mdev = 1.085/2.362/4.312/1.199 ms
```

Therefore, two-way communication between the Placement-Server and Placement-Client was successfully validated. The placement application and virtualization requirements were successfully demonstrated.

Analysis and Engineering Decision

The implemented solution successfully demonstrated cloud-based placement data management and virtualization. Zoho Creator was selected because it provides a simple platform for creating and managing the placement application without requiring separate application infrastructure.

For virtualization, VMware Workstation Pro was used to create the Placement-Server and Placement-Client VMs. A Host-only network was selected because it allows direct communication between the two VMs in a controlled private network.

Advantages

- Centralized placement data management.
- Easy cloud-based access and collaboration.
- Simple VM creation and configuration.
- Controlled private network communication.
- Low-cost solution suitable for academic demonstration.

Limitations

- The solution uses limited VM resources.
- VMware Workstation Pro is mainly suitable for local virtualization rather than large-scale cloud deployment.
- The application is designed as a basic academic prototype.

Engineering Decision

Zoho Creator and VMware Workstation Pro were selected because they are simple, practical and suitable for demonstrating the required cloud, virtualization, networking, access control and collaboration concepts.

The decision was validated by the successful VM communication test: 4 packets transmitted, 4 received, and 0% packet loss in both directions.

Pseudo Code

START

1. Open the Zoho Creator Placement Management application.
2. Add student details.
3. Add company details.
4. Create placement drive details.

5. Store and manage placement results.
 6. Provide controlled access to students and placement staff.
 7. Allow users to access and share placement information through the cloud.
 8. Create Placement-Server virtual machine in VMware.
 9. Create Placement-Client virtual machine in VMware.
 10. Configure CPU, RAM, storage and network for both VMs.
 11. Assign Host-only network to both VMs.
 12. Obtain the IP address of both VMs.
 13. Test Client → Server communication using ping.
 14. Test Server → Client communication using ping.
 15. If communication is successful,
 Record the result.
 Else,
 Check the network configuration.
 16. Verify that all placement and virtualization requirements are satisfied.
- END

Broader Considerations

- **Sustainability:** The cloud-based system reduces the need for maintaining multiple physical systems and supports efficient use of computing resources.
- **Environment:** Cloud-based data management can reduce paper-based placement records and physical storage requirements.
- **Society:** The system improves access to placement information for students and helps placement staff manage activities efficiently.
- **Safety:** Controlled access helps protect placement information from unauthorized users.
- **Ethics:** Student and company information should be accessed only by authorized users and handled responsibly.
- **Economics:** Using Zoho Creator provides a simple and cost-effective solution for an academic placement management system.
- **Accessibility:** Students and placement staff can access the cloud application through a web browser from different locations.
- **Professional Responsibility:** The system demonstrates responsible use of cloud computing, virtualization, data security and network resources.

Conclusion

The proposed solution successfully developed a cloud-based university placement management system using Zoho Creator and demonstrated virtualization using VMware Workstation Pro.

The major findings are:

- Placement information can be centrally managed through Zoho Creator.
- Controlled access and cloud collaboration can be provided to students and placement staff.
- Two virtual machines, Placement-Server and Placement-Client, were successfully created and configured.
- Two-way communication between the VMs was successfully achieved with 0% packet loss.

The major requirements of cloud application development, virtualization, networking and collaboration were successfully demonstrated.

Limitations

- The system is a basic academic prototype.
- VM resources are limited.
- Advanced security and large-scale deployment were not implemented.

Possible Improvements

- Add stronger authentication and security features.
- Add email notifications for placement updates.
- Add advanced analytics and placement prediction.
- Deploy the system on a production cloud infrastructure.

Student Reflection

Through this assignment, I gained practical knowledge of cloud computing, cloud application development, virtualization and networking. I learned how to use Zoho Creator to manage placement data and provide cloud-based access and collaboration. I also learned how to create and configure Placement-Server and Placement-Client virtual machines using VMware Workstation Pro.

One important challenge was establishing communication between the two VMs. Initially, the VMs could not communicate using NAT networking. I changed both VMs to a Host-only network, after which communication was successfully established with 0% packet loss in both directions.

If I had additional time and resources, I would improve the system by adding stronger authentication, email notifications, advanced placement analytics and a production-level cloud deployment. This would make the system more secure, useful and suitable for real-world university placement activities.

References

- Schiller, R.J. and Larochelle, D., 2024. Data Engineering Best Practices: Architect robust and cost-effective data solutions in the cloud era. Packt Publishing Ltd.
- Patil, M., 2025. Cloud computing for Industry 5.0. In Industry 5.0: Key technologies and drivers (pp. 119-154). Cham: Springer Nature Switzerland.
- Fajinmi, J. and Oloyede, J., 2025. Enhancing warehouse operations with IoT integrating RFID for real-time inventory tracking and cloud-based data management.
- Panda, S.P. and Padhy, A., 2025. Business intelligence with Power BI and Tableau: Cloud-based data warehousing, predictive analytics, and artificial intelligence-driven decision support. Deep Science Publishing.
- Sajja, J.W., Reddy, B.R.T. and Puram, S.K., 2026. Enterprise Data Transformation through Cloud Migration Architecture and Governance. Data Processing and Networking: Proceedings of ICDPN 2025, Volume 1, 1, p.444.
- Upadhyaya, P. and Koka, H., 2026. The Role of Cloud-Based BI Systems in Enabling Real-Time Data Analytics. In Architecting Strategic Business Transformation Through Data-Driven Systems (pp. 247-284). IGI Global Scientific Publishing.